diamonds, we expect that sooner or later a chance fluctuation within this highly energetic, turbulent arena of the primordial universe will cause the inflaton field's value to jump to the correct, uniform value in some small nugget of space, initiating an outward burst of inflationary expansion. As explained in the previous section, calculations show that the nugget of space need only have been tiny—on the order of 10^{-26} centimeters across—for the ensuing cosmological expansion (inflationary expansion followed by standard big bang expansion) to have stretched it larger than the universe we see today. Thus, rather than assuming or simply declaring that conditions in the early universe were right for inflationary expansion to take place, in this way of thinking about things an ultramicroscopic fluctuation weighing a mere twenty pounds, occurring within an ordinary, unremarkable environment of disorder, gave rise to the necessary conditions.

What's more, just as the slot machine will also generate a wide variety of nonwinning results, in other regions of primordial space other kinds of inflaton fluctuations would also have happened. In most, the fluctuation wouldn't have had the right value or have been sufficiently uniform for inflationary expansion to occur. (Even in a region that's a mere 10^{-26} centimeters across, a field's value can vary wildly.) But all that matters to us is that there was one nugget that yielded the space-smoothing inflationary burst that provided the first link in the low-entropy chain, ultimately leading to our familiar cosmos. As we see only our one big universe, we only need the cosmic slot machine to pay out once.[5]

Since we are tracing the universe back to a statistical fluctuation from primordial chaos, this explanation for time's arrow shares certain features with Boltzmann's original proposal. Remember from Chapter 6 that Boltzmann suggested that everything we now see arose as a rare but every so often expectable fluctuation from total disorder. The problem with Boltzmann's original formulation, though, was that it could not explain why the chance fluctuation had gone so far overboard and produced a universe hugely more ordered than it would need to be even to support life as we know it. Why is the universe so vast, having billions and billions of galaxies, each with billions and billions of stars, when it could have drastically cut corners by having, say, just a few galaxies, or even only one?